

Name: _____ Date: _____

Period: _____

Keep in Mind

This is a practice test. It mirrors the real Unit 8 test in length, format, and the ideas it covers, but the numbers and contexts differ. Work every problem with no notes, then check your answers against the key.

Part A — Vocabulary (8 pts)

Write the letter of the best description next to each term.

- | | |
|------------------------------------|---|
| 1. ReLU _____ | (A) The building block $\text{ReLU}(A\mathbf{v} + \mathbf{b})$: multiply by weights, add a bias, then rectify. |
| 2. Neural network layer _____ | (B) The step size s in gradient descent — too small crawls, too big overshoots. |
| 3. Piecewise linear function _____ | (C) The rectifier $\max(0, x)$: keep positive inputs, send negatives to 0. |
| 4. Convolution filter _____ | (D) A small weight pattern slid across the input to detect one feature everywhere. |
| 5. Weight sharing _____ | (E) The rule $\mathbf{w} \leftarrow \mathbf{w} - s \nabla L$: step opposite the slope to shrink the loss. |
| 6. Gradient descent _____ | (F) A function built of straight pieces joined at folds (a sum of shifted ReLUs). |
| 7. Learning rate _____ | (G) Reusing one filter at every position, so a giant layer needs only a few weights. |
| 8. Covariance matrix _____ | (H) The symmetric matrix of variances and covariances; its eigenvectors are the principal components. |

Part B — Multiple Choice (12 pts)

Circle the best answer.

- ReLU(-3) equals (A) -3 (B) 0 (C) 3 (D) 1
- One neural-network layer computes (A) $A\mathbf{v}$ (B) $\text{ReLU}(A\mathbf{v} + \mathbf{b})$ (C) $\mathbf{v} + \mathbf{b}$ (D) $A^{-1}\mathbf{v}$
- A piecewise linear function built from N folds (kinks) has (A) $N - 1$ pieces (B) N pieces (C) $N + 1$ pieces (D) $2N$ pieces
- Sliding one small filter across the whole input is convolution; reusing that filter everywhere is (A) weight sharing (B) the bias (C) the loss (D) a rotation
- Gradient descent updates a weight by stepping (A) along the slope (B) opposite the slope (downhill) (C) to a random weight (D) to 0
- A positive covariance $\sigma_{xy} > 0$ between two features means they (A) move in opposite directions (B) tend to move together (C) are unrelated (D) are equal

Part C — Short Answer & Computation (35 pts)

Show your work.

1. Evaluate one layer $\text{ReLU}(A\mathbf{v} + \mathbf{b})$ where $A = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$, $\mathbf{v} = \begin{bmatrix} 1 \\ 3 \end{bmatrix}$, $\mathbf{b} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$. Show $A\mathbf{v}$, then $A\mathbf{v} + \mathbf{b}$, then the ReLU.
2. The tent function $F(x) = \text{ReLU}(x) - 2\text{ReLU}(x - 2) + \text{ReLU}(x - 4)$. **(a)** Find $F(0), F(2), F(4), F(6)$. **(b)** Where are its folds, and what is the peak point?
3. Slide the edge filter $\mathbf{w} = [-1, 1]$ (output = next - current) across $\mathbf{v} = [2, 2, 2, 6, 6]$. **(a)** Write the feature map. **(b)** Where is the edge, and why is the output 0 on the flat parts?
4. The filter $[-1, 1]$ on a length-4 signal is the convolution matrix $A = \begin{bmatrix} -1 & 1 & 0 & 0 \\ 0 & -1 & 1 & 0 \\ 0 & 0 & -1 & 1 \end{bmatrix}$. **(a)** Compute $A \begin{bmatrix} 1 \\ 1 \\ 6 \\ 6 \end{bmatrix}$. **(b)** How many weights does this shared filter use, versus a full 3×4 layer?
5. Gradient descent on the loss bowl $L(w) = (w - 4)^2$, with slope $L'(w) = 2w - 8$, learning rate $s = 0.25$, starting at $w_0 = 0$. Use $w \leftarrow w - s L'(w)$. **(a)** Find w_1, w_2, w_3 . **(b)** Give the loss at each step. **(c)** What weight is it heading toward, and how would you know to stop?
6. Two-weight loss $L = (w_1 - 1)^2 + (w_2 - 3)^2$ has gradient $\nabla L = (2(w_1 - 1), 2(w_2 - 3))$. Starting at $(0, 0)$ with $s = 0.25$, take **one** gradient-descent step. Where does the loss bottom out?

7. Four students' two quiz scores are $(2, 4), (4, 2), (6, 8), (8, 6)$. **(a)** Find the mean point. **(b)** Find the variances σ_x^2, σ_y^2 and the covariance σ_{xy} . **(c)** Write the covariance matrix V . **(d)** Find its eigenvalues and the fraction of the total variance the first principal component (PC_1) holds.

Part D — Extended Response (10 pts)

Explain your reasoning in full sentences.

1. A neural network that learns is built entirely from tools in this course. **(a)** Explain what a single layer $\text{ReLU}(A\mathbf{v} + \mathbf{b})$ does, and why widening it (adding more shifted ReLUs) produces a flexible piecewise-linear function that can bend to fit data. **(b)** Explain how gradient descent *chooses* the weights by rolling downhill on the squared-error loss, and why a slope of 0 marks the best weights.
2. The covariance matrix $V = \begin{bmatrix} \sigma_x^2 & \sigma_{xy} \\ \sigma_{xy} & \sigma_y^2 \end{bmatrix}$. **(a)** Explain why V is symmetric, and why that guarantees real eigenvalues and perpendicular eigenvectors. **(b)** Explain what those eigenvectors and eigenvalues tell you about the data cloud, and why the total variance equals the trace = the sum of the eigenvalues.